

Population : Simply, population means the aggregate of human individuals in a defined area or region. In statistics, population refers the totality of all the items or individuals having some specific characteristics. e.g., All the students of university of Dhaka constitute a population.

A population can be classified into one of two groups-

(a) **Finite population :** A population having a finite number of units (or individuals or items) is called a finite population.

Example : The population consisting the students of Dhaka University.

(b) **Infinite population :** A population having an infinite number of units (or individuals or items) is called an infinite population.

Example : The population consisting of all possible outcomes (here head and tail) in successive tosses of a coin.

Sample : A representative and considerably small part of a population is known as a sample of that population.

Example :

(i) A group of 1050 students from 31,000 students of Dhaka University constitute a sample.

(ii) A spoonful boiling rice from a cooking pot of boiling rice.

2.02 : Variable

The characteristic which varies over the units (or from unit to unit) is called variable.

Example : Father's occupation of the students of Dhaka University. Here, the unit or population-unit is a student and their father's occupation varies. Say, one father's occupation is teaching, another father's occupation is business, else other father's occupation is driving and so on.

2.03 : Classification of variable

We can classify the variables according to the characteristics into two groups :

(a) Qualitative variable

(b) Quantitative variable

(a) Qualitative variable : There are many characteristics that can not be expressed in any numerical form (i. g., color, sex, occupation etc) but we can arrange them according to their quality or attribute. These types of characteristic are known as qualitative variable.

Example : Father's occupation of the students of Dhaka University.

(b) Quantitative variable : The characteristics of an unit or item that are expressed in numerical form or in numbers are called quantitative variable.

Example : Monthly family income of the students of Dhaka University. Quantitative variable can further be classified into two sub-groups :

(i) Discrete variable

(ii) Continuous variable

(i) Discrete variable

A quantitative variable which possesses isolated or integral value is called discrete variable.

Example : Family size, population size. Number of road accidents per day in Bangladesh.

(ii) Continuous variable

A quantitative variable which takes value within a range or limit is called continuous variable.

Example : Height, weight etc.

2.04 : Difference between Qualitative and Quantitative Variable

Both the qualitative and quantitative variables vary, but there lies some difference between them in respect of measurement. Some differences are :

(i) The qualitative variable measures the qualitative characteristics whereas the quantitative variable measures the quantitative characteristics.

(ii) We classify the values of quantitative variable according to numerical value, but the values of qualitative variable are classified according to quality or attribute.

(iii) We can calculate all the measures of central tendency for quantitative variable. And for qualitative variable only the median and mode can be calculated.

(iv) The frequency distribution of a qualitative variable is discrete and for quantitative variable the frequency distribution may be either discrete or continuous.

2.05 : Difference between Discrete and Continuous Variable

Although both the discrete and continuous variables are expressed in numbers, there rests some differences between them. Several differences are as follows :

(i) A discrete variable possesses an isolated or integral value whereas a continuous variable takes value within a range.

(ii) The values of a discrete variable are exact but the values of a continuous variable is approximate.

(iii) The values are integer normally in discrete variable and the values can be either integer or fraction in continuous variable.

(iv) For our easier analysis we may arrange the discrete variables as continuous. But we can't do the same for any continuous variable.

2.06 : Constant

A numerical characteristic which does never change or vary its value is termed as constant.

Example : The ratio of circumference and diameter of a circle (π). And the value of $\pi = 3.1416$ is same for all the circles.

2.07 : Scale of Measurements

To measure a variable there are various ways. We classify the ways into four scales of measurement.

- (a) Nominal scale
- (b) Ordinal scale
- (c) Interval scale
- (d) Ratio scale

(a) Nominal Scale : A scale that measures a variable nominally (or by name without any order) is called nominal scale. Here, we can set numerical values for the names but can not order them meaningfully. By setting values we only can categorize them. The variable which is measured in nominal scale is known as nominal variable.

Example : Religion, color, sex.

The nominal scale has the following properties :

- (i) Data categories are mutually exclusive and exhaustive.
- (ii) Data categories have no logical order.

(b) Ordinal Scale : A scale that identifies the values of a variable and arranges the values meaningfully in order of magnitude is called ordinal scale. Here, the magnitude is used only for comparison and not for any mathematical operation. The variable which is measured in nominal scale is known as ordinal variable, e.g., economic status. We can order the economic status in high, middle and low as follows :

Low = 1, middle class = 2, higher class = 3.

The properties of ordinal scale are :

- (i) The data classifications are mutually exclusive and exhaustive.
- (ii) Data classifications are ranked or ordered according to the particular trait they possess.

(c) Interval Scale : A scale which includes a definition of distance between the categories in terms of fixed and equal units is called interval scale. In this scale the concept of zero (0) is not included. So we can not perform any arithmetical or mathematical operation. The variable which is measured in interval scale is known as interval variable, e.g., temperature, I. Q. score. Here, thermometer records temperature in terms of degrees and a 1 degree change (increase or decrease) in temperature implies the same amount of heat.

The properties of interval scale are :

- (i) Data classifications are mutually exclusive and exhaustive.
- (ii) Data classifications are scaled according to the amount of the characteristic they possess.

(iii) Equal differences in the characteristic are represented by equal differences in the measurements.

(d) Ratio Scale : The scale of measurement which includes all the properties of an interval scale including the concept of zero (0) is called ratio scale. In ratio scale we can perform the arithmetic operation (i.e, summation, subtraction etc.). The variable that is measured in ratio scale is known as ratio variable, e.g., height, weight etc.

The properties of the ratio scale are :

- (i) Data classifications are mutually exclusive and exhaustive.
- (ii) Data classifications are scaled according to the amount of the characteristics they possess.
- (iii) Equal differences in the characteristic are represented by equal differences in the numbers assigned to the classifications.
- (iv) The zero point is the absence of the characteristic.

The following chart summerizes the major characteristics of the various levels (scale) of measurement.



